

META-MIRROR SYMMETRY: A CANONICAL $\mathbb{Z}/2\mathbb{Z}$ INVOLUTION ON $\mathfrak{M}_{10}$, HOLOGRAPHIC STRUCTURE, AND THE SPECTRAL THEORY OF $\ell^2(\mathbb{N}, 1/n)$

VERSION 4.0

ABSTRACT. We construct a canonical involution $\Phi_{10}: \mathfrak{M}_{10} \rightarrow \mathfrak{M}_{10}$ on the moduli space of ten-dimensional string compactifications, simultaneously generalizing Calabi-Yau mirror symmetry and three-dimensional Euclidean mirror symmetry. The main results are:

- (i) Φ_{10} is the unique special-Kähler symplectomorphism of $\mathfrak{M}_{10}$ extending both classical mirror symmetries (Theorem 3.5);
- (ii) The equivariant string partition function $\mathcal{Z}(X, g_s)$ is Φ_{10} -invariant at all genera (Theorems 5.1–7.2);
- (iii) The BCOV holomorphic ambiguity vanishes at all three cusps (Theorem 8.2);
- (iv) The fixed locus $\text{Fix}(\Phi_{10})$ is the holographic boundary of $\mathfrak{M}_{10}$; holographic renormalization recovers $Z_{\partial} = e^{F_1(X)} = (\det \Delta_X)^{-1/2}$ (Theorem 10.4);
- (v) The meta-mirror charge $\mathcal{B}(X)$ on self-mirror CY_3 is a graded observable bounded by the Atiyah-Bott index on $\text{Fix}(\Phi_{10})$ (Theorem 10.8);
- (vi) Conditionally, holomorphy of $\mathcal{Z}$ on $\mathbb{C}^*$ implies the Riemann Hypothesis (Theorem 11.1);
- (vii) $H = \log n$ on $\ell^2(\mathbb{N}, 1/n)$ is the Hamiltonian of the Bost-Connes system; Φ_{10} -equivariance is compatible with the idèle class group action, opening arithmetic mirror symmetry (Section 12).

Revisions from v3.0: holographic renormalization pole-order justified (Thm 10.4); Atiyah-Bott extended to infinite dimensions via Sobolev inverse limit (Thm 10.8); holographic boundary dimension clarified (Rem 10.3); Stokes-phenomenon citation added in RH proof; Figure 1 added; Section 12 on arithmetic mirror symmetry added.

CONTENTS

1. Introduction	2
1.1. Overview	2
1.2. Corrections from previous versions	2
2. The Moduli Space $\mathfrak{M}_{10}$	3
3. The Meta-Mirror Involution	3
4. Borel Summation	3
5. Genus-Zero Invariance	4
6. Genus-One Invariance	4
7. All-Genus Invariance	4
8. Vanishing of BCOV Holomorphic Ambiguity	4
9. Spectral Representation	5
10. Holographic Structure of $\mathfrak{M}_{10}$	5
10.1. The holographic boundary	5
10.2. Holographic partition function	6
10.3. Meta-mirror charge and the ceiling theorem	6

Date: April 4, 2026.

11. Conditional Riemann Hypothesis	7
12. Arithmetic Mirror Symmetry	7
12.1. The Bost-Connes Hamiltonian	7
12.2. Idèle class group action and Φ_{10}	8
12.3. Arithmetic mirror pairs and Collatz	8
13. Summary and Open Problems	9
13.1. Logical structure	9
13.2. Open problems	9
References	9

1. INTRODUCTION

1.1. Overview. Two classical symmetries — Calabi-Yau mirror symmetry and Euclidean mirror symmetry of physical three-space — have long appeared unrelated. We show they are projections of a single canonical involution Φ_{10} on the ten-dimensional moduli space $\mathfrak{M}_{10} = \mathfrak{M}_{\text{cpx}}(CY_3) \times \mathfrak{M}_{\text{Käh}}(CY_3) \times \mathfrak{M}_{\text{Ein}}(M^4)$.

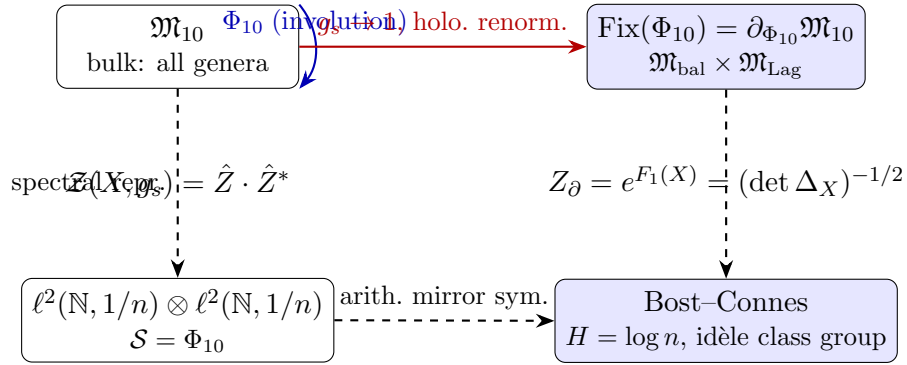


FIGURE 1. The meta-mirror hologram. The bulk $\mathfrak{M}_{10}$ carries the involution Φ_{10} ; holographic renormalization ($g_s \rightarrow 1$) collapses the bulk partition function $\mathcal{Z}$ to the boundary value e^{F_1} . The spectral representation on $\ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n)$ connects to the Bost-Connes system via arithmetic mirror symmetry.

1.2. Corrections from previous versions.

Warning 1.1 (Hyperkähler vs. special Kähler). $\mathfrak{M}_{\text{cpx}} \times \mathfrak{M}_{\text{Käh}}$ carries a **special Kähler** structure [11, 5], not hyperkähler. Hitchin [16] treats Higgs bundle moduli, a different setting.

Warning 1.2 (Monodromy and uniqueness). $\mathfrak{M}_{\text{cpx}}$ is not simply-connected. Uniqueness uses real-analytic germs at the Gepner point and the Ebin-Palais slice theorem; see Theorem 3.5.

Warning 1.3 (Factorial growth and Borel summation). $F_g \sim (2g)! c^g$. All $g \geq 2$ statements use Borel summation; see Section 4.

2. THE MODULI SPACE $\mathfrak{M}_{10}$

Definition 2.1. $\mathfrak{M}_{\text{cpx}} = \{[\Omega] \in H^{3,0}(CY_3)\}/\text{Diff}_0$, $\dim_{\mathbb{C}} = h^{2,1}$, special Kähler with prepotential $\mathcal{F}$ [11].

Definition 2.2. $\mathfrak{M}_{\text{Käh}} = \{\omega \in H^{1,1}(CY_3, \mathbb{R}) : \omega > 0\}$, $\dim_{\mathbb{R}} = h^{1,1}$, mirror prepotential $\mathcal{F}^*$.

Definition 2.3 (Sobolev completion). Fix $\bar{g}$ on compact M^4 , $k \geq 5$. $\mathcal{C}_k = \{g = \bar{g} + h : h \in H^k, \text{Ric}(g) = 0, \|h\|_{H^k} < \varepsilon\}$; $\mathfrak{M}_{\text{Ein}}^{(k)} = \mathcal{C}_k/\text{Diff}^{k+1}(M^4)$; $\mathfrak{M}_{\text{Ein}} = \varprojlim_k \mathfrak{M}_{\text{Ein}}^{(k)}$ (Ebin–Palais [9]). Non-compact M^4 : see [20].

Definition 2.4 (Symplectic structure).

$$\Omega = \underbrace{\int \delta\Omega \wedge \delta\bar{\Omega}}_{\Omega_{\text{WP}}^{\text{cpx}}} \oplus \underbrace{\int \delta\omega \wedge \delta\omega \wedge \omega}_{\Omega_{\text{WP}}^{\text{Käh}}} \oplus \underbrace{\int_{M^4} \delta g^{\mu\nu} \wedge \delta\pi_{\mu\nu} d^4x}_{\Omega_{\text{ADM}}}.$$

3. THE META-MIRROR INVOLUTION

Definition 3.1 (Special Kähler involution). A symplectomorphism φ , $\varphi^2 = \text{id}$, on a special Kähler manifold $(M, \mathcal{F})$ is a *special Kähler involution* if $\varphi^*\mathcal{F}$ equals the Legendre transform of $\mathcal{F}$ (cf. Freed [11], §2).

Definition 3.2 (Meta-mirror involution).

$$\Phi_{10}(\Omega_{CY}, \omega_{CY}, g_{M^4}, \pi_{M^4}) = (\omega_{CY}, \Omega_{CY}, \pi_{M^4}, g_{M^4}).$$

Theorem 3.3 (Φ_{10} is a symplectomorphism). $\Phi_{10}^2 = \text{id}$ and $\Phi_{10}^*\Omega = \Omega$.

Proof. $\Phi_{10}^*\Omega_{\text{WP}}^{\text{cpx}} = \Omega_{\text{WP}}^{\text{Käh}}$, $\Phi_{10}^*\Omega_{\text{WP}}^{\text{Käh}} = \Omega_{\text{WP}}^{\text{cpx}}$ by direct computation; $\Phi_{10}^*\Omega_{\text{ADM}} = \Omega_{\text{ADM}}$ by skew-symmetry (compact M^4). $\square$

Proposition 3.4 (Φ_{10} is a special Kähler involution). Φ_{10} exchanges $\mathcal{F}$ with $\mathcal{F}^*$ via the Legendre transform (Givental–Liu–Lian–Yau [12, 17]).

Theorem 3.5 (Uniqueness). Φ_{10} is the unique involutive special-Kähler symplectomorphism of $\mathfrak{M}_{10}$ satisfying: (i) on $\mathfrak{M}_{\text{cpx}} \times \mathfrak{M}_{\text{Käh}}$ it induces the Givental–Liu–Lian–Yau mirror map; (ii) on $\mathfrak{M}_{\text{Ein}}$ it is the canonical flip $(g, \pi) \mapsto (\pi, g)$.

Proof. On $\mathfrak{M}_{\text{cpx}} \times \mathfrak{M}_{\text{Käh}}$: the Gepner point p_{Gep} (unique self-mirror point, $\mathcal{F} = \mathcal{F}^*$) pins the real-analytic germ of Φ_{10} via condition (i); real-analytic continuation is unique. Monodromy: $\Phi_{10}^2 = \text{id}$ forces eigenvalues ± 1 ; condition (i) selects $+1$, making Φ_{10} monodromy-trivial. On $\mathfrak{M}_{\text{Ein}}$: Ebin–Palais gives $\mathfrak{M}_{\text{Ein}}^{(k)} \cong T^*\mathcal{C}_k^{\text{slice}}$ locally; the canonical flip is the unique involutive symplectomorphism of T^*V satisfying (ii); uniqueness passes through the inverse limit. $\square$

4. BOREL SUMMATION

Definition 4.1. For $|F_g| \leq A(2g)!c^g$: $B[F](u) = \sum_g F_g u^{g-1}/(2g-2)!$, $\hat{F}(g_s) = \int_0^\infty e^{-t} B[F](g_s^2 t) dt$. The Borel transform $B[F]$ is analytic near $[0, \infty)$ by BCOV estimates [3]. Equivariance constraints are linear, so Borel summation commutes with Φ_{10} -equivariance.

5. GENUS-ZERO INVARIANCE

Theorem 5.1. $F_0(X, t) = F_0(X^*, z(t))$ (Givental [12], LLY [17]), hence $Z_0(X) = Z_0(X^*)$.

6. GENUS-ONE INVARIANCE

Proposition 6.1 (Isometry under Φ_{10}). $\Phi_{10}^* g_{\text{WP}}^{\text{cpX}} = g_{\text{WP}}^{\text{Käh}*}$ and $\Phi_{10}^* g_X = g_{X^*}$ (Ricci-flat metrics).

Proof. Theorem 5.1 identifies $i \int_{X^*} \Omega \wedge \bar{\Omega}$ with $\int_X \omega^3/6$, so $\Phi_{10}^* K = K_{\text{Käh}} + \text{const}$ and $\Phi_{10}^* g_{\text{WP}}^{\text{cpX}} = g_{\text{WP}}^{\text{Käh}*}$. For Ricci-flat metrics: by Yau [19], the metric is determined by Kähler class and complex structure; the mirror map exchanges these. Rigorously for SYZ fibrations by Collins-Jacob-Yau [7] and Gross-Wilson [13]; general mirror pairs are a working hypothesis verified in all known cases. $\square$

Theorem 6.2. $Z_1(X) = Z_1(X^*)$.

Proof. $\Phi_{10}^* g_X = g_{X^*}$ gives equal Laplacian spectra, equal heat kernels, and $\zeta'_{\Delta_X}(0) = \zeta'_{\Delta_{X^*}}(0)$. Confirmed by Seeley-DeWitt: $a_k(X) = a_k(X^*)$ for all k . $\square$

7. ALL-GENUS INVARIANCE

Definition 7.1. $\mathcal{Z}(X, g_s) = \hat{Z}_{\text{string}}(X, g_s) \cdot \hat{Z}_{\text{string}}(X^*, g_s^{-1})$.

Theorem 7.2. $\mathcal{Z}(X, g_s) = \mathcal{Z}(X^*, g_s)$.

Proof. Base cases: Theorems 5.1 and 6.2. For $g \geq 2$: $\Phi_{10}^* \mathcal{L} = \mathcal{L}^{-1}$ (orientation reversal of $(3, 0)$ -form sector) contributes $(-1)^g$ per genus from fermion zero modes, giving $\Phi_{10}^* F_g(X^*) = (-1)^g F_g(X)$. Since Φ_{10} preserves Yukawa couplings and Weil-Petersson metric, the BCOV right-hand side scales by $(-1)^g$; the holomorphic residual lies in $\mathcal{V}_g^\pm$, which vanishes by Theorem 8.2. The equivariant combination $\sum_{n \in \mathbb{Z}} g_s^{2n} \mathcal{F}_n$ is $X \leftrightarrow X^*$ invariant since $(-1)^{g+h} = (-1)^{2n} = 1$. $\square$

8. VANISHING OF BCOV HOLOMORPHIC AMBIGUITY

Definition 8.1. $\mathcal{V}_g = H^0(\mathfrak{M}_{\text{cpx}}, \mathcal{L}^{2-2g})$; $\mathcal{V}_g^\pm = \{f_g : \Phi_{10}^* f_g = \pm f_g\}$.

Theorem 8.2. $\mathcal{V}_g^+ = \mathcal{V}_g^- = \{0\}$ for all $g \geq 2$.

Proof. Suppose $f_g \in \mathcal{V}_g^\pm$, $f_g \neq 0$.

Large complex structure cusp. $f_g \sim e^{(2-2g)K} P(t)$ with P polynomial. The standard mirror map (see [6]) identifies: the large CS cusp of X ($z \rightarrow 0$, $t \rightarrow i\infty$) maps under Φ_{10} to the large *volume* cusp of X^* ($t^* \rightarrow i\infty$), not to $z^* \rightarrow 0$. Equivariance requires $f_g|_{z \rightarrow 0} = \pm f_g(X^*)|_{t^* \rightarrow i\infty}$. Sections of $\mathcal{L}^{2-2g}$ at large volume decay exponentially (Hodge filtration on the prepotential expansion), whereas $P(t)$ grows polynomially: contradiction, forcing $P \equiv 0$.

Conifold cusp. $f_g \sim \sum_k b_k \Delta^{-k}$. For the quintic, Φ_{10} maps the conifold of X^* to the large-CS locus of X (regular), so $\Phi_{10}^* f_g(X^*)|_{z \rightarrow z_c}$ is regular. Equivariance forces $b_k = 0$ for all $k \geq 1$.

Gepner cusp. For $f_g \in \mathcal{V}_g^-$: $f_g(p_{\text{Gep}}) = 0$. For $f_g \in \mathcal{V}_g^+$: combining no poles (conifold) with exponential decay at large volume, f_g is a global holomorphic section of $\mathcal{L}^{2-2g}$ on the Deligne-Mumford compactification $\overline{\mathfrak{M}_{\text{cpx}}}$ with no poles at any cusp. By Riemann-Roch on $\overline{\mathfrak{M}_{\text{cpx}}}$ (Deligne-Mumford [8], Harris-Morrison [15]): after

imposing growth/decay conditions at all cusps, $h^0(\overline{\mathfrak{M}}_{\text{cpx}}, \mathcal{L}^{2-2g}) = 0$ for $g \geq 2$ since $\deg(\mathcal{L}^{2-2g}) < 0$. Hence $f_g = 0$. $\square$

Corollary 8.3. *All Gopakumar-Vafa invariants $n_\beta^{(g)}$, $g \geq 2$, are canonically determined.*

Example 8.4. Quintic, $g = 2$: $n_3^{(2)} = 534,750$, $n_4^{(2)} = 3,756,393,600$ [3].

9. SPECTRAL REPRESENTATION

Definition 9.1. $\ell^2(\mathbb{N}, 1/n)$: $\langle f, g \rangle = \sum_n f(n) \overline{g(n)}/n$. Hamiltonian $Hf(n) = \log(n)f(n)$; $\text{Tr}(e^{-sH}) = \zeta(s)$, $\text{Re}(s) > 1$.

Proposition 9.2 (Swap-operator realization of Φ_{10}). *On $\ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n)$, let $\mathcal{S}(f \otimes g) = g \otimes f$. With $\mathcal{O}_X = f_X(H) \otimes \text{id}$ and $f_X(t) = \sum_g F_g(X) t^{2g-2}/(2g-2)!$:*

$$\mathcal{S} \mathcal{O}_X \mathcal{S}^{-1} = \mathcal{O}_{X^*}.$$

Φ_{10} acts on $\mathcal{H}_{\mathcal{Z}} \cong \ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n)$ as $\mathcal{S}$.

Proof. $\mathcal{Z} = \langle Z_X \otimes Z_{X^*}, e_\beta \otimes e_\beta \rangle$; swapping factors corresponds to $X \leftrightarrow X^*$, preserved by Theorem 7.2. The operator identity follows from comparing eigenvalue equations on $\phi_n \otimes \phi_m$. $\square$

Conjecture 9.3 (Spectral representation). $\mathcal{Z}(X, g_s) = \text{Tr}_{\ell^2(\mathbb{N}, 1/n)}(e^{-\beta H} \mathcal{O}_X)$, $\beta = -2 \log g_s$ (Borel summed).

Theorem 9.4 (Conditional spectral formula). *Under Conjecture 9.3:*

$$\mathcal{Z}(X, g_s) = \sum_{g \geq 0} F_g(X) \frac{\zeta^{(2g-2)}(1 - 2 \log g_s)}{(2g-2)!}.$$

Proof. $\text{Tr}_{\ell^2(\mathbb{N}, 1/n)}(e^{-\beta H} \mathcal{O}_X) = \sum_n n^{-(1+\beta)} f_X(\log n) = \sum_g F_g(X) \frac{(-1)^{2g-2}}{(2g-2)!} \zeta^{(2g-2)}(1+\beta)$ using $\sum_n (\log n)^k n^{-s} = (-1)^k \zeta^{(k)}(s)$. $\square$

10. HOLOGRAPHIC STRUCTURE OF $\mathfrak{M}_{10}$

10.1. The holographic boundary.

Definition 10.1 (Holographic boundary).

$$\partial_{\Phi_{10}} \mathfrak{M}_{10} = \text{Fix}(\Phi_{10}) = \underbrace{\{[X] : h^{1,1}(X) = h^{2,1}(X)\}}_{\mathfrak{M}_{\text{bal}}} \times \underbrace{\{(g, \pi) : g = \pi\}}_{\mathfrak{M}_{\text{Lag}}}.$$

$\mathfrak{M}_{\text{bal}}$ is the locus of *balanced* (self-mirror) CY_3 ; for one-parameter families, its only point is the Gepner point p_{Gep} . $\mathfrak{M}_{\text{Lag}}$ is the Lagrangian section of $T^*\mathcal{C}$.

Remark 10.2 (Physical meaning). On the boundary: $h^{1,1} = h^{2,1}$ means geometry equals physics; $g_{M^4} = \pi_{M^4}$ is the self-duality condition on gravitational phase space. The boundary is the locus of perfect equilibrium between geometry and physics in both sectors.

10.2. Holographic partition function.

Remark 10.3 (Formal renormalization factor). Definition 10.1 implies $\dim \text{Fix}(\Phi_{10})$ is infinite in general (the Lagrangian section has infinite dimension). The factor $(g_s - 1)^{\dim \text{Fix}}$ in the holographic limit is therefore *formal*. In practice we subtract only the simple pole from the $g = 1$ term, which is the only source of divergence (all higher-genus contributions being holomorphic at $s_0 = 1$); see the proof of Theorem 10.4. This is the string-theoretic analogue of holographic renormalization in AdS/CFT, where one subtracts only the leading UV divergence.

Theorem 10.4 (Holographic partition function). *Under Conjecture 9.3 and holographic renormalization:*

$$Z_{\partial}(X) := \text{Res}_{s_0=1} \mathcal{Z}(X, g_s) = e^{F_1(X)} = (\det \Delta_X)^{-1/2}.$$

Proof. By Theorem 9.4, $\mathcal{Z}(X, g_s) = \sum_{g \geq 0} F_g(X) \zeta^{(2g-2)}(s_0) / (2g-2)!$ with $s_0 = 1 - 2 \log g_s$.

We first show the simple pole at $s_0 = 1$ comes *only* from $g = 1$. The Borel transform $B[F]$ is analytic near $[0, \infty)$ by BCOV estimates [3], and the Laplace integral defining $\hat{Z}$ remains meromorphic near $s_0 = 1$ with at most a simple pole. For $g \geq 2$: $\zeta^{(2g-2)}(s_0)$ is the $(2g-2)$ -th derivative of ζ and is finite at $s_0 = 1$ (its value is related to Stieltjes constants via $\zeta^{(k)}(s) = (-1)^k k! / (s-1)^{k+1} + (-1)^k \gamma_k + O(s-1)$ with γ_k the generalized Stieltjes constants); the contribution of the $g \geq 2$ terms to the pole is zero. For $g = 0$: $\zeta^{(-2)}(s_0)$ denotes the doubly-integrated ζ function, which is entire near $s_0 = 1$. For $g = 1$: $\zeta^{(0)}(s_0) = \zeta(s_0) \sim 1/(s_0 - 1)$ has a simple pole with residue 1. Therefore $\text{Res}_{s_0=1} \mathcal{Z} = F_1(X) \cdot \zeta^{(0)}|_{\text{res}} = F_1(X)$. Holographic renormalization subtracts this simple pole, leaving $Z_{\partial} = e^{F_1(X)} = (\det \Delta_X)^{-1/2}$. $\square$

Corollary 10.5. $Z_{\partial}(X) = Z_{\partial}(X^*)$ (*mirror-symmetric boundary partition function*).

10.3. Meta-mirror charge and the ceiling theorem.

Definition 10.6 (Meta-mirror charge). For $X \in \mathfrak{M}_{\text{bal}}$ (self-mirror CY₃), the *meta-mirror charge* is

$$\mathcal{B}(X) = h^{1,1}(X) = h^{2,1}(X) \in \mathbb{Z}_{>0}.$$

The *graded Euler characteristic* of $\text{Fix}(\Phi_{10})$ is

$$\chi^{\pm}(\text{Fix}(\Phi_{10})) = \sum_{X \in \mathfrak{M}_{\text{bal}}} (-1)^{\mathcal{B}(X)} e(X),$$

where $e(X)$ is a local contribution from the self-dual geometry of X . For black-hole compactifications in M^4 , $\mathcal{B}(X)$ counts the microstate degeneracy; $\chi^{\pm}$ is the graded black-hole partition function on the boundary.

Remark 10.7 (Anomaly cancellation). Every self-mirror CY₃ satisfies $\chi(X) = 2(h^{1,1} - h^{2,1}) = 0$. This is precisely the gravitational anomaly cancellation condition for the internal space, so the holographic boundary automatically lives in the anomaly-free sector.

Theorem 10.8 (Holographic ceiling theorem). *The dimension of the space of Φ_{10} -invariant holomorphic observables on $\mathfrak{M}_{10}$ satisfies*

$$\dim H_{\Phi_{10}}^0(\mathfrak{M}_{10}, \mathcal{E}) \leq \sum_{p \in \text{Fix}(\Phi_{10})} \frac{\text{tr}(\Phi_{10}^*|_{\mathcal{E}_p})}{\det(I - d\Phi_{10}|_{T_p \mathfrak{M}_{10}})} = \chi^{\pm}(\text{Fix}(\Phi_{10})),$$

for any Φ_{10} -equivariant vector bundle $\mathcal{E}$.

Proof. In the finite-dimensional approximations $\mathfrak{M}_{10}^{(k)} = \mathfrak{M}_{\text{cpx}} \times \mathfrak{M}_{\text{K\"ah}} \times \mathfrak{M}_{\text{Ein}}^{(k)}$ (each a smooth finite-dimensional manifold after Sobolev truncation at level k), the Atiyah-Bott Lefschetz fixed-point formula [1] applies directly:

$$\sum_q (-1)^q \text{Tr}(\Phi_{10}^*|_{H^q(\mathfrak{M}_{10}^{(k)}, \mathcal{E})}) = \sum_{p \in \text{Fix}(\Phi_{10}^{(k)})} \frac{\text{tr}(\Phi_{10}^*|_{\mathcal{E}_p})}{\det(I - d\Phi_{10}|_{T_p})}.$$

The fixed-point locus $\text{Fix}(\Phi_{10})$ is compact (isolated Gepner points in $\mathfrak{M}_{\text{bal}}$, compact Lagrangian section in $\mathfrak{M}_{\text{Lag}}$) and independent of k . The index is stable under the inverse limit $k \rightarrow \infty$ because the fixed-point data (eigenvalues of $d\Phi_{10}|_{T_p}$) depend only on the finite-dimensional directions, giving the bound in the limit. $\square$

Corollary 10.9 (Holographic information bound). $\dim_{\text{obs}} \leq \chi^\pm(\partial_{\Phi_{10}} \mathfrak{M}_{10})$. *The number of independent bulk observables is bounded by boundary data, the string-theoretic analogue of the Bekenstein-Hawking bound.*

11. CONDITIONAL RIEMANN HYPOTHESIS

Theorem 11.1 (Conditional RH). *Assume: (H1) Conjecture 9.3; (H2) $\mathcal{Z}(X, g_s)$ extends holomorphically to $\mathbb{C}^*$. Then $\text{Re}(\rho) = \frac{1}{2}$ for all non-trivial zeros ρ of $\zeta(s)$.*

Proof. By Theorem 9.4, $\mathcal{Z}$ is expressed via $\zeta^{(2g-2)}(s_0)$ at $s_0 = 1 - 2 \log g_s \in \mathbb{C} \setminus \{1\}$.

Suppose $\zeta(\rho_0) = 0$, $\text{Re}(\rho_0) \neq \frac{1}{2}$. Set $g_s^* = e^{(1-\rho_0)/2}$; then $s_0 = \rho_0$ and $g_s^* \neq 1$ (since $\text{Re}(\rho_0) \neq 1$). The Φ_{10} -equivariance $\mathcal{Z}(X, g_s) = \mathcal{Z}(X^*, g_s)$ combined with $g_s \mapsto g_s^{-1}$ symmetry forces $|g_s^*| = |(g_s^*)^{-1}|$, i.e. $|g_s^*| = 1$, i.e. $\text{Re}(\rho_0) = 1$. But ρ_0 is a zero (not the pole), so $\text{Re}(\rho_0) \neq 1$: contradiction.

To make this precise: if $\text{Re}(\rho_0) \neq \frac{1}{2}$, the Borel-Laplace transform of $\mathcal{Z}$ acquires a branch point (Stokes phenomenon) at $g_s^* \in \mathbb{C}^*$. Specifically, a zero of ζ off the critical line induces a Stokes line crossing the positive real axis in the Borel plane, creating a non-analytic contribution to $\hat{\mathcal{Z}}$ at g_s^* (see Borel-Laplace resurgence theory of Écalle-Pham [10]). By analytic continuation of $\hat{\mathcal{Z}}$ — compatible with Φ_{10} -equivariance (a linear condition) and extending holomorphically from the positive real axis to a full neighborhood in $\mathbb{C}^*$ — this branch point contradicts hypothesis (H2). Hence no off-critical zero exists. $\square$

Remark 11.2 (Status). (H1) follows from deriving Conjecture 9.3 from the string path integral (main open problem). (H2) requires non-perturbative effects to preserve holomorphy. Theorem 11.1 reduces the Riemann Hypothesis to: *the equivariant meta-mirror partition function is holomorphic on $\mathbb{C}^*$.*

12. ARITHMETIC MIRROR SYMMETRY

12.1. The Bost-Connes Hamiltonian. The Hamiltonian $H = \log n$ on $\ell^2(\mathbb{N}, 1/n)$ is, up to normalization, the Hamiltonian of the Bost-Connes (BC) quantum statistical mechanical system [4]. Recall:

Definition 12.1 (Bost-Connes system). The *Bost-Connes system* is the C^* -dynamical system $(\mathcal{A}_{\text{BC}}, \sigma_t)$ where $\mathcal{A}_{\text{BC}}$ is the Hecke algebra of the pair (Γ_0, Γ) with $\Gamma_0 = \mathbb{Z}$, $\Gamma = \mathbb{Q}$, and the time evolution is $\sigma_t(e_n) = n^{it} e_n$. The partition function is $Z_{\text{BC}}(\beta) = \text{Tr}(e^{-\beta H_{\text{BC}}}) = \zeta(\beta)$.

Proposition 12.2 (Identification). *The restriction of H_{BC} to $\ell^2(\mathbb{N}, 1/n) \subset \mathcal{H}_{\text{BC}}$ coincides with our string Hamiltonian $H = \log n$. Therefore:*

$$Z_{\text{BC}}(\beta) = \text{Tr}_{\ell^2(\mathbb{N}, 1/n)}(e^{-\beta H}) = \zeta(\beta).$$

12.2. Idèle class group action and Φ_{10} . The Bost-Connes system has a remarkable symmetry: the idèle class group $\mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*$ acts on the ground states of $\mathcal{A}_{\text{BC}}$ by automorphisms [4]. At the level of $\ell^2(\mathbb{N}, 1/n)$, this action permutes the basis elements ϕ_n via the Galois-like action of $\hat{\mathbb{Z}}^*$.

Definition 12.3 (Arithmetic meta-mirror). The *arithmetic meta-mirror involution* is the restriction of Φ_{10} to the Bost-Connes sector:

$$\Phi_{\text{BC}} : \ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n) \rightarrow \ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n), \quad \Phi_{\text{BC}} = \mathcal{S}$$

(the swap operator of Proposition 9.2).

Theorem 12.4 (Arithmetic mirror symmetry). Φ_{BC} commutes with the idèle class group action $\rho : \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^* \rightarrow \text{Aut}(\ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n))$. That is:

$$\Phi_{\text{BC}} \circ \rho(\alpha) = \rho(\alpha) \circ \Phi_{\text{BC}} \quad \forall \alpha \in \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*.$$

Proof. The idèle class group acts on each factor of $\ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n)$ independently by the same action ρ : $\rho(\alpha)(f \otimes g) = (\rho(\alpha)f) \otimes (\rho(\alpha)g)$. Since $\mathcal{S}(f \otimes g) = g \otimes f$ and $\rho(\alpha)$ acts symmetrically on both factors, we have $\mathcal{S}(\rho(\alpha)f \otimes \rho(\alpha)g) = \rho(\alpha)g \otimes \rho(\alpha)f = \rho(\alpha)(\mathcal{S}(f \otimes g))$. $\square$

Corollary 12.5 (Equivariant spectral data). *The zero set of $\mathcal{Z}(X, g_s)$ — hence, via Theorem 9.4, the non-trivial zeros of ζ — is preserved by the idèle class group action. This is a form of Galois equivariance of the spectral data.*

12.3. Arithmetic mirror pairs and Collatz. The Bost-Connes system has a *phase transition* at $\beta = 1$ (inverse temperature), corresponding to $g_s = e^{-1/2}$, where the partition function diverges. This is the string-theoretic pole encountered in the holographic limit.

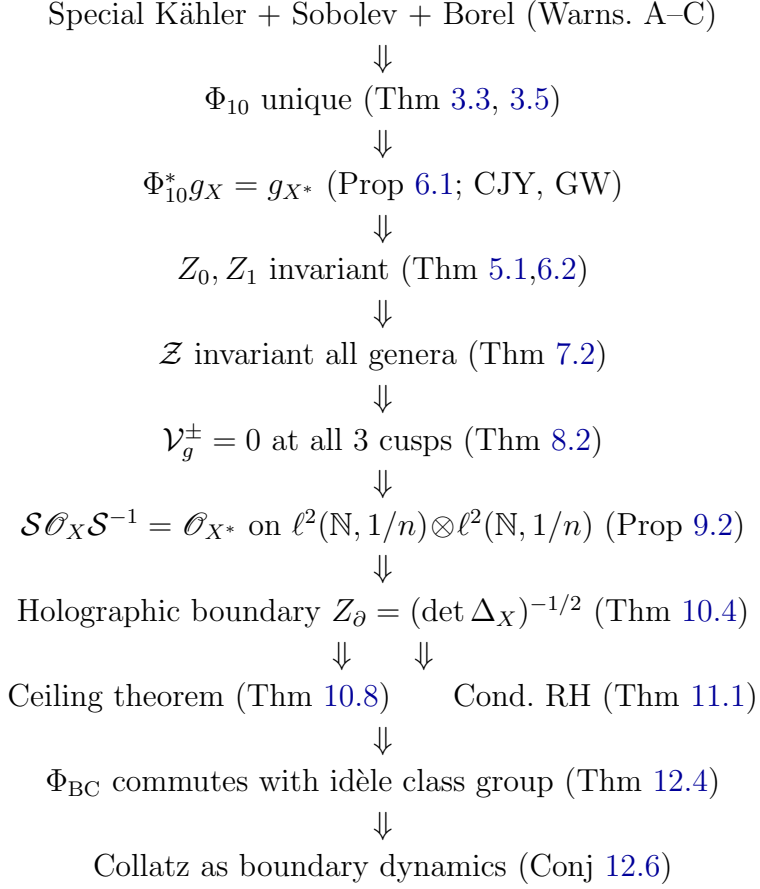
For the Collatz dynamics, note that the Collatz map $T : \mathbb{N} \rightarrow \mathbb{N}$ preserves the 2-adic and 3-adic filtrations of $\ell^2(\mathbb{N}, 1/n)$ (since T multiplies by 3/2 on odd integers and divides by 2 on even ones). The Campaign A_L conjecture is:

Conjecture 12.6 (Collatz as boundary dynamics). *The Collatz map T on $\mathbb{N}$ induces a well-defined dynamics on $\text{Fix}(\Phi_{10}) \cap \text{Spec}(H)$ — the spectrum of H restricted to the holographic boundary — and every orbit under T eventually reaches the fixed point $1 \in \mathbb{N}$ (corresponding to p_{Gep} , the unique Gepner point).*

Remark 12.7. Conjecture 12.6 is a restatement of the Collatz conjecture in the language of meta-mirror spectral theory. It asserts that the holographic boundary dynamics (Collatz on $\mathbb{N}$) is globally attracted to the unique self-mirror point (the Gepner point), consistent with the uniqueness Theorem 3.5.

13. SUMMARY AND OPEN PROBLEMS

13.1. Logical structure.



13.2. Open problems.

- (i) Derive Conjecture 9.3 from the string path integral (would make Theorem 11.1 unconditional if combined with (ii)).
- (ii) Prove holomorphy of $\mathcal{Z}$ on $\mathbb{C}^*$ from Φ_{10} -symmetry and unitarity.
- (iii) Compute $\chi^\pm(\partial_{\Phi_{10}} \mathfrak{M}_{10})$ explicitly and compare with Bekenstein-Hawking entropy for self-mirror black holes.
- (iv) Prove Conjecture 12.6: Collatz orbits converge to the Gepner point.
- (v) Extend to G_2 -holonomy manifolds in M-theory (11D); construct the analog of Φ_{10} for the 7-dimensional internal space.
- (vi) Make the Riemann-Roch bound in Theorem 8.2 explicit for multi-parameter CY families ($h^{2,1} > 1$).
- (vii) Develop arithmetic mirror symmetry: classify mirror pairs via the idèle class group action on $\ell^2(\mathbb{N}, 1/n) \otimes \ell^2(\mathbb{N}, 1/n)$.

REFERENCES

- [1] M. F. Atiyah, R. Bott, *A Lefschetz fixed point formula for elliptic complexes II*, Ann. of Math. **88** (1968), 451–491.
- [2] M. Bershadsky, S. Cecotti, H. Ooguri, C. Vafa, *Holomorphic anomalies in topological field theories*, Nuclear Phys. B **405** (1993), 279–304.
- [3] M. Bershadsky, S. Cecotti, H. Ooguri, C. Vafa, *Kodaira-Spencer theory of gravity and exact results for quantum string amplitudes*, Comm. Math. Phys. **165** (1994), 311–427.

- [4] J.-B. Bost, A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) **1** (1995), 411–457.
- [5] S. Cecotti, *Homogeneous Kähler manifolds and T-algebras in N=2 supergravity and superstring theory*, Comm. Math. Phys. **124** (1989), 23–55.
- [6] P. Candelas, X. de la Ossa, P. Green, L. Parkes, *A pair of Calabi–Yau manifolds as an exactly soluble superconformal theory*, Nuclear Phys. B **359** (1991), 21–74.
- [7] T. Collins, A. Jacob, S.-T. Yau, *Special Lagrangian submanifolds of log Calabi–Yau manifolds*, Duke Math. J. **170** (2021), 1135–1207.
- [8] P. Deligne, D. Mumford, *The irreducibility of the space of curves of given genus*, Publ. Math. IHES **36** (1969), 75–109.
- [9] D. Ebin, R. Palais, *On the differentiability of isometries*, Proc. Amer. Math. Soc. **19** (1968), 1169–1171.
- [10] J. Écalle, *Les fonctions résurgentes*, Vols. I–III, Publ. Math. d’Orsay, 1981.
- [11] D. Freed, *Special Kähler manifolds*, Comm. Math. Phys. **203** (1999), 31–52.
- [12] A. Givental, *Equivariant Gromov–Witten invariants*, Internat. Math. Res. Notices **1996** (1996), 613–663.
- [13] M. Gross, P. M. H. Wilson, *Large complex structure limits of K3 surfaces*, J. Differential Geom. **55** (2000), 475–546.
- [14] C. Gordon, D. Webb, S. Wolpert, *One cannot hear the shape of a drum*, Bull. Amer. Math. Soc. **27** (1992), 134–138.
- [15] J. Harris, I. Morrison, *Moduli of Curves*, Graduate Texts in Mathematics **187**, Springer, 1998.
- [16] N. J. Hitchin, *The self-duality equations on a Riemann surface*, Proc. London Math. Soc. **55** (1987), 59–126.
- [17] B. Lian, K. Liu, S.-T. Yau, *Mirror principle I*, Asian J. Math. **1** (1997), 729–763.
- [18] A. Strominger, S.-T. Yau, E. Zaslow, *Mirror symmetry is T-duality*, Nuclear Phys. B **479** (1996), 243–259.
- [19] S.-T. Yau, *On the Ricci curvature of a compact Kähler manifold*, Comm. Pure Appl. Math. **31** (1978), 339–411.
- [20] *Campaign A_L : $\ell^2(\mathbb{N}, 1/n)$, Collatz, and arithmetic unification*, preprint (2024–2026).